Online Appendix to "Access beyond ownership: Stock Connect closures and price delay in Chinese stocks"

Table OA1. Alternative holdings measures, winsorization, and return frequency.

	Eligible	LogHold	TopQuint	NoP99	Raw	Trim05	Five
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. DelayIF							
Treatment	3.097***	2.724***	3.936***	0.990***	0.640***	0.637***	0.462***
	(5.85)	(5.61)	(5.38)	(5.73)	(5.87)	(5.86)	(5.58)
Wild p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
Panel B. DelayIH							
Treatment	2.947***	2.686***	3.967***	0.977***	0.637***	0.634***	0.420***
	(5.93)	(5.77)	(5.72)	(5.78)	(5.94)	(5.93)	(4.96)
Wild p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
Stock_FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date_FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs	2,212,912	2,207,616	2,201,883	2,184,558	2,207,616	2,207,616	2,203,242

Notes. Each observation is a stock-day in the closure episodes and their control days, and every column uses 45 episodes. *Treatment* is *Closed* interacted with the holdings measure. Column (1), *Eligible*, uses the eligibility indicator, column (2), *LogHold*, the log of one plus *Hold*, column (3), *TopQuint*, an indicator for the top quintile of *Hold* within the episode relative to the bottom quintile, with the other quintiles included, and column (4), *NoP*99, *Hold* after dropping stock-episodes above the 99th percentile of *Hold*. Column (5), *Raw*, uses outcomes without winsorization, column (6), *Trim*05, winsorizes at the 0.5th and 99.5th percentiles, and column (7), *Five*, computes the delay from five-minute returns with five lags. *Wild p* is the wild cluster bootstrap p-value with clustering by episode and 9,999 draws. The t-statistics in parentheses are calculated using standard errors clustered by stock and by date. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table OA2. Placebo dates.

	Holiday		Back20		Back40	
	DelayIF	DelayIH	DelayIF	DelayIH	DelayIF	DelayIH
	(1)	(2)	(3)	(4)	(5)	(6)
Placebo×Hold	0.013	0.077	-0.094	-0.165	0.051	0.074
	(0.05)	(0.33)	(-0.52)	(-0.85)	(0.30)	(0.36)
Wild p	0.942	0.691	0.314	0.108	0.614	0.605
Stock_FE	Yes	Yes	Yes	Yes	Yes	Yes
Date_FE	Yes	Yes	Yes	Yes	Yes	Yes
Episodes	14	14	45	45	45	45
Obs	790,765	790,765	869,064	869,064	745,344	745,344

Notes. Each observation is a stock-day on placebo dates and their control days. In columns (1) and (2), *Holiday*, the placebo dates are 30 normal trading days, in 14 episodes, next to holidays shared by the mainland and Hong Kong. In columns (3) to (6), *Back*20 and *Back*40, each closure episode is shifted back by 20 or 40 trading days to dates on which northbound trading is open. *Placebo* is one on a placebo date, and the level of *Hold* is included but not

reported. *Wild p* is the wild cluster bootstrap p-value with clustering by episode and 9,999 draws. The t-statistics in parentheses are calculated using standard errors clustered by stock and by date. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table OA3. Quarterly holdings and subperiods.

	PreDaily	PreQuart	AllQuart	Post	PostAug16
	(1)	(2)	(3)	(4)	(5)
Panel A. DelayIF					
Closed×Hold	0.745***	0.739***	0.627***	0.341	0.248
	(6.61)	(6.55)	(5.81)	(1.02)	(0.69)
Wild p	<0.001	<0.001	<0.001	0.267	0.428
MDE	0.317	0.317	0.303	0.946	1.019
Panel B. DelayIH					
Closed×Hold	0.701***	0.694***	0.624***	0.363	0.306
	(6.47)	(6.54)	(5.96)	(1.04)	(0.86)
Wild p	<0.001	<0.001	<0.001	0.322	0.408
MDE	0.304	0.298	0.294	0.987	1.000
Stock_FE	Yes	Yes	Yes	Yes	Yes
Date_FE	Yes	Yes	Yes	Yes	Yes
Obs	1,653,182	1,633,086	2,187,344	554,434	532,607
Episodes	35	35	45	10	10

Notes. Each observation is a stock-day in the closure episodes and their control days. Columns (1) and (2) use the 35 episodes before August 2024, with *Hold* from the last daily disclosure (*PreDaily*) or from the last quarter-end disclosure (*PreQuart*) before the episode. Column (3), *AllQuart*, uses all episodes with quarter-end holdings. Columns (4) and (5) use the ten episodes after August 2024, with the latest quarter-end holdings (*Post*) or the last daily holdings of 16 August 2024 (*PostAug*16). *MDE* is the minimum detectable effect at 5% significance and 80% power. *Wild p* is the wild cluster bootstrap p-value with clustering by episode and 9,999 draws. The t-statistics in parentheses are calculated using standard errors clustered by stock and by date. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table OA4. Weather closures and permanent loss of buying access.

	FullDay		HalfDay		SellOnly	
	DelayIF	DelayIH	DelayIF	DelayIH	DelayIF	DelayIH
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	0.852**	0.879**	0.798**	0.573	1.294*	1.054
	(2.09)	(2.03)	(2.41)	(1.37)	(1.76)	(1.48)
Wild p	0.067	0.040	0.054	0.239	0.248	0.412
Events	6	6	6	6	14	14
Obs	256,939	256,939	1,186,151	1,186,151	46,830	46,830

Notes. In columns (1) and (2), *FullDay*, each observation is a stock-day around six full-day closures of northbound trading caused by typhoon or black-rainstorm warnings, and *Treatment* is *Closed*×*Hold* with stock-fixed effects and date-fixed effects. In columns (3) and (4), *HalfDay*, each observation is a stock-segment on six days on which northbound trading closed for part of the day, and *Treatment* compares the closed and open segments of the same stock and day, interacted with *Hold*. In columns (5) and (6), *SellOnly*, each observation is a stock-day around 14 batches of index reviews, the treated stocks lose northbound buying

access permanently but can still be sold, the control stocks leave the same indices but remain buyable, and *Treatment* is one after the change for treated stocks, with stock-batch and batch-date fixed effects. *Events* is the number of closure events or review batches; in one batch only one treated stock has holdings data, so 13 batches identify the effect. *Wild p* is the wild cluster bootstrap p-value with clustering by event or batch, using the six-point weights of Webb and 9,999 draws. Estimated batch by batch, the binary treatment effect on delay is positive in 10 of 13 batches for *DelayIF* and in 12 for *DelayIH*. The t-statistics in parentheses are calculated using standard errors clustered by stock and by date. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table OA5. Size, index membership, and analyst coverage, step by step.

	Size	Seven	Eight	EightHK	EligOnly	Decile
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. DelayIF						
Closed×Hold	0.358***	0.258***	0.238***	0.232***	0.352***	0.314***
	(5.26)	(3.99)	(3.93)	(3.87)	(4.71)	(6.66)
Wild p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
R^2	0.027	0.113	0.113	0.114	0.007	0.003
Panel B. DelayIH						
Closed×Hold	0.306***	0.253***	0.231***	0.216***	0.365***	0.255***
	(4.57)	(3.79)	(3.60)	(3.46)	(4.75)	(5.61)
Wild p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
R^2	0.018	0.079	0.080	0.080	0.005	0.002
Stock_FE	Yes	Yes	Yes	Yes	Yes	Yes
Date_FE	Yes	Yes	Yes	Yes	Yes	No
Decile×Date_FE	No	No	No	No	No	Yes
Obs	2,206,013	2,138,357	2,138,357	2,127,264	1,030,034	2,201,883
Episodes	45	44	44	44	45	45

Notes. Each observation is a stock-day in the closure episodes and their control days. Column (1), *Size*, adds *Closed* interacted with log float market value. Column (2), *Seven*, adds *Closed* interacted with log float market value, log trading value, membership in the CSI 300, SSE 50, and CSI 500 indices, the beta on IF, and the pre-closure delay. Column (3), *Eight*, adds analyst coverage to these seven, and column (4), *EightHK*, also adds the eight Hong Kong exposure measures. Column (5), *EligOnly*, uses eligible stocks only, and column (6), *Decile*, replaces *Date_FE* with fixed effects for each size decile and date (*Decile*×*Date_FE*). All characteristics are measured before the episode, and their levels and interactions are included but not reported. *Wild p* is the wild cluster bootstrap p-value with clustering by episode and 9,999 draws. The t-statistics in parentheses are calculated using standard errors clustered by stock and by date. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table OA6. Market activity on closure days and quiet-day placebos.

	DelayIF	DelayIH	Sync
	(1)	(2)	(3)
Panel A. Closed×Hold with controls for market activity			
State, no controls	0.609***	0.574***	-0.145***
	(5.78)	(5.62)	(-4.60)
Wild p	<0.001	<0.001	0.001
State, controls	0.215***	0.181**	0.009

	(3.04)	(2.53)	(0.34)
Wild p	0.012	0.030	0.752
Nonzero minutes	0.507***	0.546***	-0.188***
	(4.28)	(5.03)	(-5.96)
Wild p	<0.001	<0.001	<0.001
Panel B. Quiet days as placebo events			
Quiet×Hold, both 40%	0.050	0.230	-0.043
	(0.38)	(1.58)	(-0.77)
Wild p	0.759	0.174	0.572
Closed−Quiet, both 40%	0.584***	0.400**	-0.183***
	(3.48)	(2.28)	(-2.87)
Wild p	0.002	0.055	0.026
Quiet×Hold, IF variance 30%	0.467***	0.606***	-0.270***
	(8.03)	(9.49)	(-12.76)
Wild p	<0.001	<0.001	<0.001
Closed−Quiet, IF variance 30%	0.152	0.011	0.049
	(1.27)	(0.10)	(1.32)
Wild p	0.328	0.939	0.259
Stock_FE	Yes	Yes	Yes
Date_FE	Yes	Yes	Yes

Notes. Each observation is a stock-day. In Panel A, *State, no controls* adds the IF realized variance and the log market trading value of the day, each interacted with *Hold* and with *Closed×Hold*, to the main regression without controls (2,207,616 stock-days, 45 episodes), and the coefficient is *Closed×Hold* at the mean of the control days. *State, controls* adds the same terms to the regression with the sixteen characteristics of Table 3, Panel B (2,127,264 stock-days, 44 episodes). *Nonzero minutes* adds the number of minutes with nonzero returns and its interaction with *Closed* to the regression without controls. In Panel B, quiet days are normal northbound trading days on which both market trading value and IF realized variance are in the bottom 40% of their earlier distribution (85 days in 47 episodes), or on which IF realized variance alone is in the bottom 30% (544 days in 224 episodes). Closure days and quiet days enter one regression with common stock-fixed effects and date-fixed effects; *Quiet×Hold* is the quiet-day coefficient, and *Closed−Quiet* is the closure coefficient minus the quiet-day coefficient. With the bottom 30% on both variables (34 days), *Quiet×Hold* is 0.21 for *DelayIF* (wild p 0.29) and *Closed−Quiet* is 0.42 (wild p 0.07). *Wild p* is the wild cluster bootstrap p-value with clustering by episode and 9,999 draws. The t-statistics in parentheses are calculated using standard errors clustered by stock and by date. ***, **, and * denote significance at the 1%, 5%, and 10% levels.